Solutions to MP363 Problem Set 1

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P.1

This is yet another example of how an inherently quantum-mechanical quantity – the energy spectrum of a black body – can have genuinely macroscopic effects, in this case the colour of a star.

(a) What we're doing here is deriving the function that tells us the spectral radiance if we characterize radiation by wavelength rather than frequency. The key is realising that a range of frequencies $d\nu$ is not the same as a range of wavelengths $d\lambda$. In fact, since $\nu = \frac{c}{\lambda}$, we see a change in frequency $d\nu$ is related to the change in wavelength by

$$d\nu = \frac{d\nu}{d\lambda}\, d\lambda = \frac{d}{d\lambda}\left(\frac{c}{\lambda}\right) d\lambda = -\frac{c}{\lambda^2}\, d\lambda .$$

The negative sign indicates that a positive change in frequency results in a negative change in wavelength: higher frequency $\Rightarrow$ shorter wavelength and vice versa. This is why we need the minus sign in $B(\nu,T)\,d\nu = -f(\lambda,T)\,d\lambda$. That gives $d\nu$ in terms of λ; to get the rest, we simply put $\nu = c/\lambda$ into B, i.e.

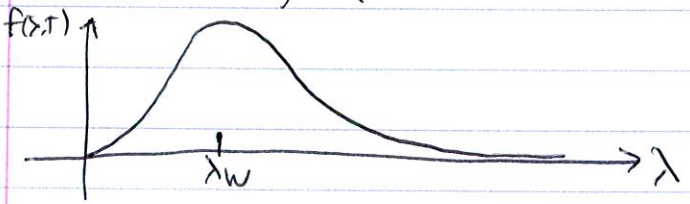
$$-f(\lambda,T)\,d\lambda = \frac{2h(c/\lambda)^3}{c^2}\,\frac{1}{\exp\left(\frac{h}{k_BT}\left(\frac{c}{\lambda}\right)\right)-1}\left(-\frac{c}{\lambda^2}\,d\lambda\right)$$

$$= -\frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_BT}-1}\,d\lambda$$

so since this holds for all wavelength intervals $d\lambda$,

$$\boxed{f(\lambda,T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_BT}-1}}$$

(b) If we plotted this as a function of λ (for a fixed T), we'd find it looks something like:



The point at which it peaks is the λ_W that we're after, so we just need to find when the function is at its maximum.

Since we're considering T to be fixed, this just means we want

$$\left.\frac{\partial f}{\partial \lambda}(\lambda,T)\right|_{\lambda=\lambda_W} = 0 .$$

The derivative is easy to find:

$$\frac{\partial f(\lambda,T)}{\partial\lambda} = 2hc^2 \frac{\partial}{\partial\lambda}\left(\frac{1}{\lambda^5(e^{hc/\lambda k_BT}-1)}\right)$$

$$= 2hc^2\left[-\frac{5\lambda^4(e^{hc/\lambda k_BT}-1)+\lambda^5\left(-\frac{hc}{\lambda^2 k_BT}e^{\frac{hc}{\lambda k_BT}}\right)}{\lambda^{10}(e^{hc/\lambda k_BT}-1)^2}\right]$$

$$= \frac{2hc^2}{\lambda^7}\frac{5\lambda+\frac{hc}{k_BT}e^{\frac{hc}{\lambda k_BT}}-5\lambda e^{\frac{hc}{\lambda k_BT}}}{(e^{\frac{hc}{\lambda k_BT}}-1)^2}$$

This vanishes at $\lambda=\lambda_w$, when the numerator is zero, i.e.

$$5\lambda_w+\frac{hc}{k_BT}e^{\frac{hc}{\lambda_w k_BT}}-5\lambda_w e^{\frac{hc}{\lambda_w k_BT}}=0.$$

Letting $x = \frac{hc}{\lambda_w k_B T}$, this becomes

$$\frac{hc}{k_BT}\left(\frac{5}{x}+e^x-\frac{5}{x}e^x\right)=0$$

Dividing out the $\frac{hc}{k_BT}$ and multiplying by xe^{-x} gives

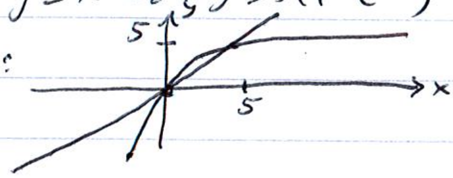
$$5e^{-x}+x-5=0$$

or

$$\boxed{x=5(1-e^{-x})}$$

as desired.

(c) You can use some sort of zero-finding program to determine what value of x satisfies this, or we could use a graphical approach: if I plot the functions $y=x$ and $y=5(1-e^{-x})$ on the same set of axes, I get:



The only two places these curves cross are at $x=0$ and a value slightly less than 5. $x=0 \Rightarrow \lambda_w$ infinite, so that's not the root we want. The other one — at approximately 4.965 — will give us the result:

$$x \approx 4.965 = \frac{hc}{\lambda_w k_B T} \Rightarrow \lambda_w T = \frac{hc}{4.965\,k_B}.$$

Plugging in the values given for h, c and k_B, we get

$$\boxed{\lambda_w T = 2.9\times10^{-3}\,\mathrm{m\cdot K}}$$

as desired.

(d) Now for an actual calculation: $T = 5000K$ gives

$$\lambda_W = \frac{2.9\times10^{-3}\,m\cdot K}{5000K} = 5.81\times10^{-7}m$$

$$= 581\ nm.$$

If I look at

https://science-edu.larc.nasa.gov/EDDOCS/Wavelengths_for_Colors.html

I find that this wavelength is smack dab in the middle of what our eyes see as $\boxed{\text{yellow}}$.

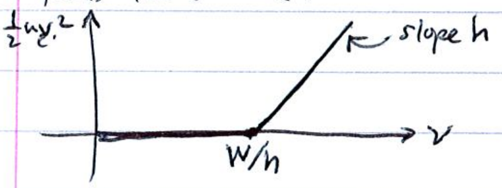
... of comparison, the Sun's surface temperature is about 5800K, ... corresponds to $\lambda_W = 50$

P-2

The basic facts we need to know for this problem are that photoelectric emission can only start when the photon energy is equal to or greater than the metal's work function, and that the electron kinetic energy is the difference between the two. In other words

$$\frac{1}{2}m_e v^2 = h\nu - W$$

when $h\nu \geq W$ and zero otherwise. A plot of the electron KE thus looks like:



(a) Here we have two points on the line above, ($\nu_1 = 2\times10^{15}Hz$, $\frac{1}{2}m_e v_1^2 = 3.9\,eV$) and ($\nu_2 = 10^{16}Hz$, $\frac{1}{2}m_e v_2^2 = 37.0\,eV$). Thus, h and W_{Ti} must values such that

$$3.9\,eV = h\nu_1 - W, \quad \frac{1}{2}m_e v_2^2 = h\nu_2 - W$$

These two simultaneous equations can be easily solved for both h and W:

$$h = \frac{\frac{1}{2}m_e v_2^2 - \frac{1}{2}m_e v_1^2}{\nu_2 - \nu_1}$$

$$W_{Ti} = \frac{\frac{1}{2}m_e v_2^2 \nu_1 - \frac{1}{2}m_e v_1^2 \nu_2}{\nu_2 - \nu_1}$$

Putting in the values given for ν_1, ν_2, $\frac{1}{2}m_e v_1^2$ and $\frac{1}{2}m_e v_2^2$ immediately gives

$$\boxed{W_{Ti} = 4.38\,eV}$$

for the titanium work function, and $h = 4.14\times10^{-15}\,eV.s$. This is a perfectly valid way to write h, but if we want it in J.s, we multiply by $\frac{1.6\times10^{-19}J}{1eV}$ to get

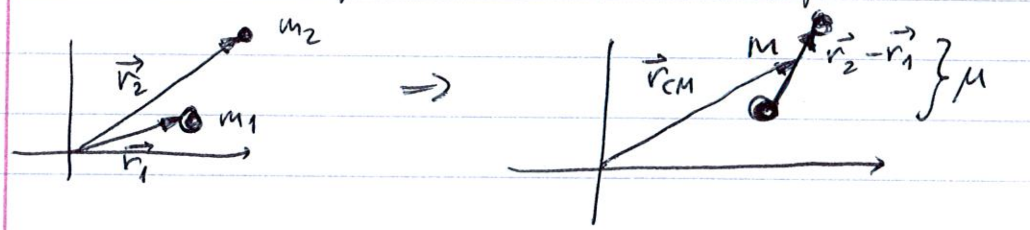
$$\boxed{h = 6.62\times10^{-34}\,J.s}$$

(b) The ν-axis intercept in the above plot is W/h, i.e. the minimum frequency necessary to start photoemission of electrons. For lithium, $W_{Li} = 2.9eV$; if we use our eV.s value of h from (a), we find

$$\nu_{min} = W_{Li}/h = 2.9eV/4.14\times10^{-15}eV.s$$
$$= \boxed{7\times10^{14}\,Hz}.$$

P.3

The key to this problem is that any two-body problem can be



reduced to one which treats the whole system as being of total mass $M = m_1+m_2$ located at its centre of mass position $\vec{r}_{CM} = \frac{m_1\vec{r_1}+m_2\vec{r_2}}{m_1+m_2}$ <u>plus</u> an internal system depending on the relative position $\vec{r} = \vec{r_1}-\vec{r_2}$, with one mass treated as infinite and the other having the reduced mass $\mu = \frac{m_1m_2}{m_1+m_2}$.

So an atom (or atom-like object) whizzes along on its centre of mass, but all internal forces like Coulomb attraction are treated independently. Thus, if we think of the electron as the attracted object and the proton/antimuon/positron as the fixed "nucleus", then the electron has an effective mass of $\mu = \frac{m_em_N}{m_e+m_N}$, where m_N is the mass of the "nuclear" particle.

So in Bohr's model, the mass he used in finding the hydrogen energy levels was $\frac{m_em_p}{m_e+m_p}$, where m_p was the proton mass. However, for these two new "atoms", we must use the mass of either the antimuon or the positron...

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(a) Muonium: $m_N = m_\mu = 207 m_e$, so $\mu = \dfrac{m_e \cdot 207 m_e}{m_e + 207 m_e} = \dfrac{207}{208} m_e$
$= 0.9952 m_e$.

But we just want to compare the energy levels to hydrogen, not find exact values: thus, if $E_n^{(H)}$ and $E_n^{(\mu)}$ are the energy levels of hydrogen and muonium, we see that

$$E_n^{(\mu)}/E_n^{(H)} = \frac{\frac{m_e m_\mu}{m_e + m_\mu}}{\frac{m_e m_p}{m_e + m_p}} = \frac{m_\mu (m_e + m_p)}{m_p (m_e + m_\mu)}$$

$$= \frac{207 m_e (m_e + 1836 m_e)}{1836 m_e (m_e + 207 m_e)}$$

$$= \frac{207 \cdot 1837}{1836 \cdot 208} = 0.9957$$

so the energy levels we'd find in muonium are $\boxed{\text{largely the same as in hydrogen,}}$ only differing by about half a percent.

(b) Positronium: $m_N = m_e$ this time, so taking the ratio of $E_n^{(p)}$ to $E_n^{(H)}$ is once again the ratio of the reduced masses:

$$E_n^{(p)}/E_n^{(H)} = \frac{\frac{m_e m_e}{m_e + m_e}}{\frac{m_p m_e}{m_p + m_e}} = \frac{1837}{2 \cdot 1836} = 0.5003.$$

In other words, the positronium energy levels are only very slightly more than $\boxed{\text{half those of hydrogen.}}$